Bai Ruofei

Singapore | bairuofei.github.io | ruofei001@e.ntu.edu.sg | 86-18867108067 | +65 94651769

Education

- **Ph.D., Robotics, Nanyang Technological University** — **Singapore**
  Advisor: Prof. Lihua Xie, Dr. Wei-Yun Yau | *GPA: 4.92/5.00* — 08/2022 - 08/2026
  - Research topics: Diffusion-based generative planning, active SLAM, robot navigation and exploration
- **Master, Control Theory and Engineering, Zhejiang University** — **China**
  Advisor: Assoc. Prof. Ronghao Zheng | *GPA: Top 3/42* — 09/2019 - 03/2022
  - Research topics: Symbolic task and motion planning under spatial-temporal constraints
- **Bachelor, Automation, Zhejiang University** — **China**
  Advisor: Assoc. Prof. Ronghao Zheng | *GPA: 3.93/4.00* — 09/2015 - 06/2019
  - Research topics: Distributed optimization and coordination for multi-robot systems

Research Project

My research focuses on robot planning and perception, generative models, and reinforcement learning.

- **Diffusion-based Generative Robot Planning | IROS 2026, Submitted to CoRL 2026** — 2025 - 2026
  Generate multi-robot coordinated trajectories with one single model and reinforcement learning fine-tuning
  - Proposed novel concept of robot tokens (Roken) to unify multi-robot trajectory generation into one diffusion transformer model, with versatile capabilities for single and multi-robot planning, and conditional planning.
  - Developed trajectory-level reinforcement learning to fine-tune the pre-trained navigation foundation model; Achieved better long-term planning capability and safety adherence than the pre-trained model.
  - Zero-shot real-world deployment of one to four mobile robots using one Roken model with good scalability; Achieved strong generalization to up to 10 robots and in partially observed cluttered environments.
- **Multi-Robot Navigation under Connectivity Constraints | ICRA 2025, Submitted to RA-L** — 2024 - 2025
  Maintain mutual communication and line-of-sight connectivity in multi-robot mapless navigation
  - Established the first Line-of-Sight constrained Multi-Robot (MR) navigation framework in unknown environments (previous methods mostly assume known maps) based on ego-centric visible regions from raw LiDAR scans.
  - Developed a graph Laplacian-based approach for connectivity maintenance with adaptive topology optimization; Achieved reliable team connectivity maintenance and efficient goal navigation in unknown environments.
  - Simulation verification in large-scale MR-exploration tasks (Gazebo), and real-world deployment with three Tello UAVs in cluttered environments without outside localization, demonstrating superior convenience for deployment.
- **Active Localization and Mapping in Robot Exploration | RA-L 2024, IROS 2024** — 2022 - 2024
  Ensure efficient exploration and reliable localization & mapping by exploiting prior environmental information
  - Developed a hierarchical SLAM-aware exploration framework for efficient exploration with an active loop-closing mechanism; Exploited prior env. topology to predict localization uncertainty during high-level path planning.
  - Established an efficient two-stage approach with initial TSP/VRP path finding and subsequent enhancement through loop closure insertion; Achieved sub-optimality guarantee by proving submodularity property of the objective function; Implemented fast SLAM uncertainty evaluation based on abstracted pose graph topology.
  - Verified in large-scale building exploration tasks with a Pioneer3AT Robot, outperforming existing methods in localization accuracy and showing competitive efficiency.
- **Task and Motion Planning under Temporal Logic Constraints | RAS 2022, IROS 2021** — 2020 - 2022
  Transform complex temporal-logic constraints in multi-robot task planning into symbolic graph search framework
  - Established a symbolic task and path planning framework for multiple robots, transforming complex temporal-logic task constraints into symbolic graph search problems, using linear temporal logic (LTL) language formulation.
  - Developed a task decomposition method that identifies independent task sequences in mixed independent and collaborative tasks, addressing long temporal-logic constraints with significantly reduced search spaces.
  - Outperformed classical mixed-integer programming approaches in terms of feasibility and computational efficiency.
- **Selected Contributed Projects** — 2025 - Present
  Covering vision-language-navigation, vision-language-action model, and social navigation

- Humans as planners: A social navigation approach that treats humans as planners to generate socially compliant behaviors, where I developed a scoring module based on mutual visibility for leader selection. (RA-L 2025)
- High-frequency VLA: A backbone-agnostic module for VLAs to use stale semantic intent alongside real-time proprioception for high-frequency execution, where I contributed to the discussion and experiments. (IROS 2026)
- Imagination before navigation: A general-purpose VLN model based on pre-trained video generation models using inverse dynamics, grounding the common sense prior of video models to robot navigation policy. (IROS 2026)

Work Experience

- **Embodied AI Algorithm Intern** | **HUAWEI** — 07/2025 - 10/2025
  Whole-body motion planning and VLA model investigation
  - Developed an inverse-kinematics-based whole-body joint-pose solver for a mobile humanoid robot (Galaxea R1 Pro) based on Mink and MuJoCo; Enhanced end-effector manipulability for human-like natural postures.
  - Investigated latent action-based VLAs and RL for motion control; Fine-tuned SmolVLA for LiBERO benchmark.
- **Planning and Control (PnC) Algorithm Intern** | **Autowise.ai** — 04/2022 - 06/2022
  Autonomous driving path planning, trajectory optimization, and reachable set prediction
  - Implemented a series of path planning algorithms for autonomous driving, including sampling-based dynamic programming; quadratic programming with driving corridors; reachability set prediction for convex optimization.
  - Developed a patent about merging driving corridors for improved planning efficiency.
- **Team member** | **Intelligent Car Competition of ZJU** — 01/2018 - 06/2018
  Fuzzy PID control for tricycle car racing following electromagnetic lines
  - Designed segmented fuzzy PID control laws for car racing in challenging environments with sharp turns and uphills.
  - Won first prize in the University Competition (tricycle track); Extensive in-field testing for optimal performance.
- **Software Engineer Intern** | **D.Y. Innovations (Low-altitude UAV operations)** — 07/2018 - 09/2018
  Cross-platform autonomous IP address setup for local-network devices using Bonjour mDNS
  - Built a Bonjour/mDNS-based local-network discovery module for UAV devices, enabling cross-platform automatic IP address setup and broadcasting across Windows, Linux, and Android.

Publications

Preprint | **Conference** | **Journal**

[1] Robots as Tokens: Unified Diffusion Transformer for Coordinated Multi-Robot Trajectory Generation, *Submitted to CoRL 2026* [Paper] [Project] — [Arxiv] 2026
Ruofei Bai, J Chen, Y Cai, J Li, WY Yau, L Xie.

[2] IntentNav: Learning Spatial-Visual Object Navigation from Human Demonstrations, *Submitted to CoRL 2026* [Paper] [Project] — [Arxiv] 2026
Y Cai, Z Li, M Wang, M Bao, H Zhu, **Ruofei Bai**, D Zhao, Z Li, W Wang, WY Yau, J Zhang, C Lv.

[3] Line-of-Sight-Constrained Multi-Robot Mapless Navigation via Polygonal Visible Regions, *Submitted to RA-L* [Paper] [Code] — [Arxiv] 2026
Ruofei Bai, S Yuan, X Xu, X Ji, X Li, H Guo, WY Yau, L Xie.

[4] ImagiNav: Scalable Embodied Navigation via Generative Visual Prediction and Inverse Dynamics, *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [Paper] — IROS 2026
J Chen, Y Cai, Y Wang, **Ruofei Bai**, Y Cao, J Li, YW Yun, G Sartoretti.

[5] Beyond Imitation: Reinforcement Learning Fine-Tuning for Adaptive Diffusion Navigation Policies, *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [Paper] — IROS 2026
J Sheng*, **Ruofei Bai***, K Xu, R Liu, J Chen, S Yuan, WY Yau, L Xie.

[6] TIDAL: Temporally Interleaved Diffusion and Action Loop for High-Frequency VLA Control, *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [Paper] — IROS 2026
Y Sun, H Wang, **Ruofei Bai**, Z Li, J Li, M Yee, WY Yau.

[7] Gaussian Semantic Field for One-shot LiDAR Global Localization, *Submitted to RA-L* [Paper] — [Arxiv] 2025
P Yin, S Yuan, H Cao, X Ji, **Ruofei Bai**, S Chen, L Xie.

[8] Learning Generalizable Policies with Multi-Task Pre-Training for Multi-Robot Exploration, *Submitted to IJRR* — 2025

S Zhu, Y Xu, **Ruofei Bai**, L Li, J Chen, L Xie, J Xu.

[9] Realm: Real-Time Line-of-Sight Maintenance in Multi-Robot Navigation with Unknown Obstacles, *2025 IEEE International Conference on Robotics and Automation (ICRA)* [Paper] [Code] — ICRA 2025
Ruofei Bai, S Yuan, K Li, H Guo, WY Yau, L Xie. Best Conference Paper Award Finalist

[10] Swept Volume-Aware Trajectory Planning and MPC Tracking for Multi-Axle Swerve-Drive AMRs, *2025 IEEE International Conference on Robotics and Automation (ICRA)* [Paper] — ICRA 2025
T Hu, S Yuan, **Ruofei Bai**, X Xu, Y Liao, F Liu, L Xie.

[11] AirSwarm: Enabling Cost-Effective Multi-UAV Research with COTS drones, *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [Paper] [Code] — IROS 2025
X Li, K Xu, F Liu, **Ruofei Bai**, S Yuan, L Xie.

[12] Following Is All You Need: Robot Crowd Navigation Using People As Planners, *IEEE Robotics and Automation Letters (RA-L)* [Paper] [Code] — RA-L 2025
Y Liao, X Xu, **Ruofei Bai**, Y Yang, M Cao, S Yuan, L Xie.

[13] Graph-based SLAM-Aware Exploration with Prior Topo-Metric Information, *IEEE Robotics and Automation Letters (RA-L)* [Paper] [Code] — RA-L 2024
Ruofei Bai, H Guo, WY Yau, L Xie.

[14] Multi-Robot Active Graph Exploration with Reduced Pose-SLAM Uncertainty via Submodular Optimization, *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* — IROS 2024
Ruofei Bai, S Yuan, H Guo, P Yin, WY Yau, L Xie. [Paper] [Code]

[15] Hierarchical Multi-Robot Strategies Synthesis and Optimization under Individual and Collaborative Temporal Logic Specifications, *Robotics and Autonomous Systems (RAS)* [Paper] — RAS 2022
Ruofei Bai, R Zheng, M Liu, S Zhang.

[16] Multi-Robot Task Planning under Individual and Collaborative Temporal Logic Specifications, *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* [Paper] [Code] — IROS 2021
Ruofei Bai, R Zheng, M Liu, S Zhang.

Patent

[1] A Robot Temporal Task Planning Method, Apparatus, and Electronic Device, *Chinese Invention Patent, CN113408949B* — 2022
R Zheng, **Ruofei Bai**, M Liu, S Zhang.

Professional Activities

- **Teaching Assistant** — **Zhejiang University**
  - Robot Modeling and Control — 2020, 2021
  - FPGA Principles and Applications — 2020, 2021
- **Reviewer:** T-RO, RA-L, ICRA, IROS, AIM, etc

Selected Awards

- 2025 IEEE ICRA Best Conference Paper Award Finalist — 2025
- Singapore International Graduate Award (SINGA) — 2022 - 2026
- Outstanding Graduate of Zhejiang Province & ZJU (Master) | Top 5% — 2022
- Outstanding Graduate of Zhejiang Province & ZJU (Bachelor) | Top 5% — 2019
- National Encouragement Scholarship (**all three times**) — 2016, 2017, 2018
- Honorable Mention Award for 2016 MCM/ICM — 2016

Skills

Languages: Mandarin Chinese: Native; English: Professional working proficiency (CET-4,6; IELTS 7.0; GRE 322)

Programming Languages: Python, C/C++, PyTorch, MATLAB, Git, Bash, VHDL (for FPGA), Linux, Docker

Engineering Software: ROS, Gazebo, MuJoCo, Isaac Sim, Gurobi, G2O, GTSAM

Platforms: TurtleBot3, Pioneer3AT, DJI Tello UAV, MCU (Arduino, Raspberry Pi)